

Revision Notes for Class 12 Physics

Chapter 11 – Dual Nature of Radiation and Matter

11. 1 In Class 12 Physics, the chapter on "Dual Nature of Radiation and Matter" explores a fascinating concept: that light and matter can behave both as waves and as particles. This chapter introduces key ideas like the photoelectric effect, where light hitting a metal surface can release electrons, and de Broglie's hypothesis, which suggests that particles like electrons also have wave-like properties. Understanding these dual characteristics helps explain various physical phenomena and bridges classical and quantum physics. By studying this chapter, you'll gain insight into how the behaviour of light and matter can be described in different ways, leading to a deeper comprehension of the natural world.

11.2 Electron Emission:

Electron emission refers to the process where electrons are released from a material, often a metal, when it absorbs energy. This can happen in several ways, depending on the type of energy source:

1. Photoelectric Emission: When light (or photons) strikes a metal surface, it can cause the ejection of electrons. The energy of the photons must be greater than the work function (the minimum energy required to remove an electron from the metal).

The formula for this is: $E_k = h\nu - \phi$

where E_k is the kinetic energy of the emitted electron, h is Planck's constant, ν is the frequency of the incident light, and ϕ is the work function of the metal.

2. Thermionic Emission: This occurs when a metal is heated to a high temperature, causing electrons to gain enough energy to overcome the work function and escape from the metal surface.

3. Field Emission: This happens when a strong electric field is applied to a material, which can pull electrons out of the surface.

These processes illustrate how energy input can lead to the emission of electrons, demonstrating the interaction between matter and energy in different ways.

11.3 Photoelectric Effect

Hertz's observations:

Hertz's observations are key to understanding the photoelectric effect. Heinrich Hertz, a physicist, conducted experiments in the late 19th century that revealed how ultraviolet light could cause electrons to be emitted from a metal surface.

Hertz observed that when ultraviolet light struck a metal electrode, it emitted electrons, but this only happened if the light's frequency was above a certain threshold. He found that:

1. The number of emitted electrons increased with the intensity of the light.
2. The energy of the emitted electrons depended on the frequency of the light, not its intensity.

These observations were crucial in showing that light must have enough energy to free electrons from the metal surface, leading to the development of the concept of photons.

Formula:

The energy of the emitted electrons is given by: $E_k = h\nu - \phi$

Here:

E_k is the kinetic energy of the emitted electrons.

h is Planck's constant.

ν is the frequency of the incident light.

ϕ is the work function of the metal.

Hertz's work laid the foundation for the quantum theory of light and helped in understanding the dual nature of radiation.

Hallwachs' and Lenard's observations:

Hallwachs' Observation: In the late 19th century, Wilhelm Hallwachs discovered that when ultraviolet light (or any light of sufficient frequency) is directed at a metal surface, it causes the metal to emit electrons. This was the first evidence that light can cause electrons to be ejected from a metal.

Lenard's Observation: Following Hallwachs, Philipp Lenard conducted experiments that confirmed Hallwachs' findings. Lenard observed that the emitted electrons (photoelectrons) had a range of energies and their number increased with the intensity of the incident light. He also found that the kinetic energy of these electrons depended on the frequency of the light, not its intensity.

Formula:

The relationship between the kinetic energy of the emitted electrons and the light is given by the photoelectric equation:

$$E_k = h\nu - \phi$$

where:

E_k is the kinetic energy of the emitted electrons,

h is Planck's constant,

ν is the frequency of the incident light,

ϕ is the work function of the metal.

These observations were fundamental in showing that light has particle-like properties, leading to the development of quantum mechanics.

11.4 Experimental Study of Photoelectric Effect: The photoelectric effect is a key experiment in understanding the dual nature of light. In this experiment, light is shone on a metal surface, and if the light has enough energy, it ejects electrons from the metal.

1. **Setup:** A metal plate is exposed to light in a vacuum. The plate is connected to an electric circuit that measures the current of ejected electrons.

2. **Observation:** When light with a certain frequency hits the metal, electrons are emitted. The amount of current depends on the intensity of the light, while the frequency of light determines whether electrons are ejected at all.

3. **Key Findings:** The experiment shows that light behaves as particles called photons. Each photon has a specific amount of energy given by the formula: $E = h\nu$

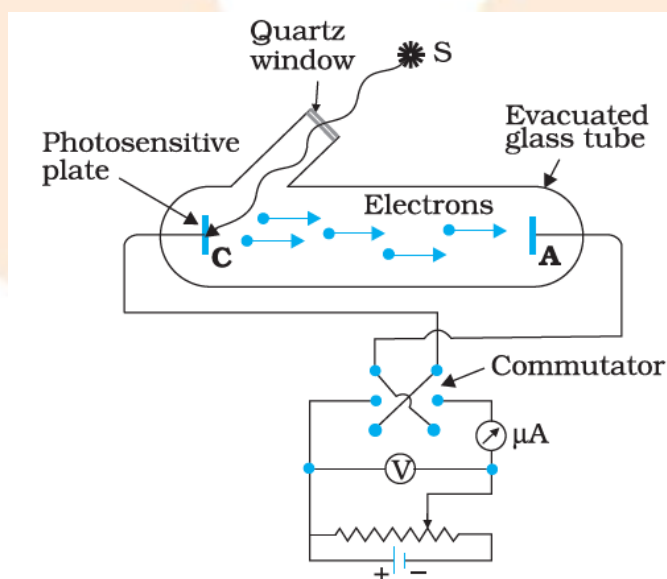
where E is the energy of the photon, h is Planck's constant, and ν (nu) is the frequency of the light.

4. **Photoelectric Equation:** The kinetic energy of the emitted electrons can be calculated using:

$$E_k = h\nu - \phi$$

where E_k is the kinetic energy of the ejected electron and ϕ is the work function of the metal (the minimum energy required to eject an electron).

This experiment supports the idea that light behaves as particles, providing evidence for the dual nature of radiation and matter.



Effect of intensity of light on Photocurrent:

Higher Intensity: If the light's intensity is increased, more photons hit the metal surface. This causes more electrons to be ejected, which increases the photocurrent.

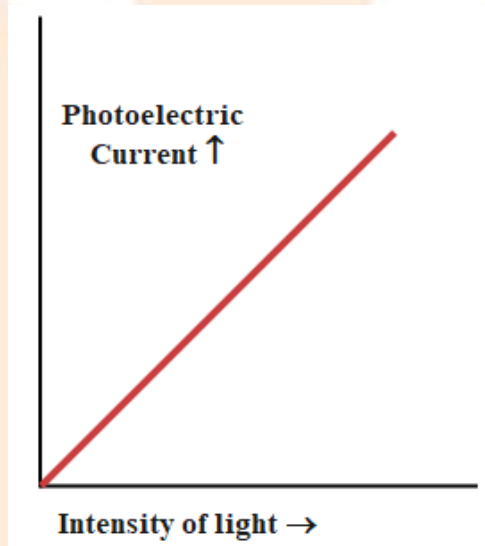
Lower Intensity: If the light's intensity is decreased, fewer photons hit the metal. This means fewer electrons are ejected, so the photocurrent is lower.

Formula:

The photocurrent (I) is directly proportional to the light intensity (I_0):

$$I \propto I_0$$

In short, more intense light produces a higher photocurrent because it ejects more electrons from the metal surface.



Effect of Potential on Photoelectric Current: It examines how changing the electric potential between the metal surface and the collector affects the current produced in the photoelectric effect.

Photoelectric Current: When light hits a metal surface, it releases electrons, creating a flow of current. This is called the photoelectric current.

Effect of Increasing Potential: If you increase the positive potential on the collector (relative to the metal surface), more emitted electrons are attracted to it, which increases the photoelectric current.

Stopping Potential: If you increase the negative potential, it can stop the electrons from reaching the collector, reducing the current to zero. The potential needed to stop the current is called the stopping potential.

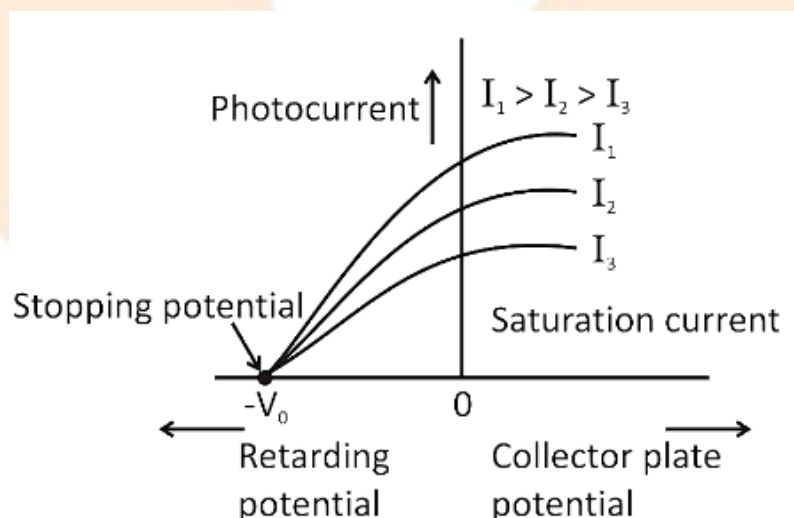
Formulas:

1. **Photoelectric Current (I):** It increases with the intensity of light and the positive potential applied.

2. **Stopping Potential (V_0):** This is related to the maximum kinetic energy of the emitted electrons: $eV_0 = E_k$

where e is the electron charge and E_k is the maximum kinetic energy of the electrons.

In summary, the photoelectric current depends on the potential applied: increasing positive potential increases the current, while increasing negative potential can reduce it to zero.



Effect of Frequency of Incident Radiation on Stopping Potential: We study how the frequency of incident radiation (light) affects the stopping potential in the photoelectric effect. When light hits a metal surface, it can release electrons. The stopping potential is the voltage needed to stop these ejected electrons from reaching the anode.

Here's how the frequency of the incident radiation affects the stopping potential:

1. Higher Frequency Increases Stopping Potential: As the frequency of the incident light increases, the energy of the photons increases. This means more energy is transferred to the electrons, making them move faster. Therefore, a higher stopping potential is required to stop these faster-moving electrons.

2. Formula to Understand the Relationship:

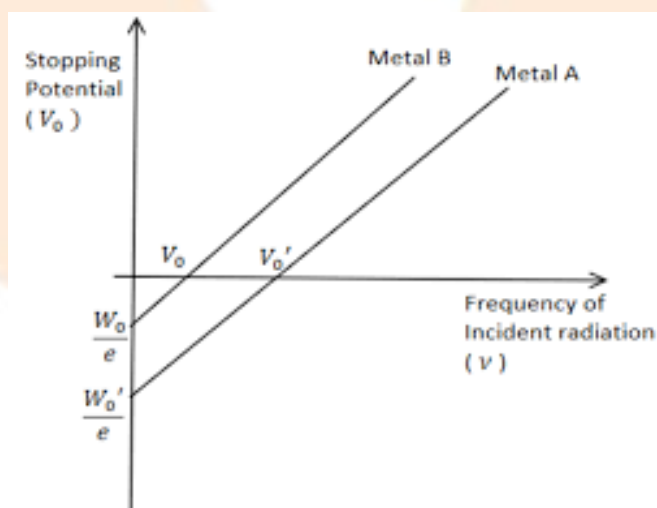
The energy of the incident photons is given by: $E = h\nu$

where h is Planck's constant and ν is the frequency of the incident radiation.

The stopping potential V_s is related to the maximum kinetic energy of the emitted electrons:

$$eV_s = h\nu - \phi$$

where e is the charge of an electron, V_s is the stopping potential, and ϕ is the work function of the metal.



11.5 Photoelectric Effect and Wave Theory of Light:

Photoelectric Effect: The photoelectric effect occurs when light shines on a metal surface and causes the release of electrons from that surface. This phenomenon shows that light behaves like particles called photons. Each photon has a specific amount of energy.

Wave Theory of Light: According to the wave theory of light, light is considered as a wave that can carry energy. However, this theory struggled to explain why electrons are emitted only when the light frequency is above a certain level, regardless of the light's intensity.

Formula:

Energy of a Photon: $E = h\nu$

Here, E is the energy of the photon, h is Planck's constant, and ν (nu) is the frequency of light.

Photoelectric Equation:

Kinetic Energy of Ejected Electrons: $E_k = h\nu - \phi$

In this equation, E_k is the kinetic energy of the emitted electrons, ϕ is the work function of the metal, and $h\nu$ is the energy of the incoming photons.

The photoelectric effect provided evidence that light can behave like particles, challenging the old wave theory and leading to the development of quantum physics.

11.6 Einstein's Photoelectric Equation: Energy Quantum of Radiation: Einstein's photoelectric equation helps us understand how light can release electrons from a metal surface. According to Einstein, light is made of particles called photons, and each photon carries a specific amount of energy.

When light hits a metal, the energy from the photons can knock electrons off the metal surface. The energy of the photon must be greater than a certain threshold, known as the work function of the metal, for this to happen.

Formula: $E_k = h\nu - \phi$

Here's what each term means:

E_k is the kinetic energy of the ejected electron.

h is Planck's constant.

ν (nu) is the frequency of the incident light.

ϕ (phi) is the work function of the metal.

This equation shows that the energy of the incoming photons (light) is used to overcome the work function, and any extra energy is converted into the kinetic energy of the ejected electrons.

11.7 Particle Nature of Light: The Photon: The "Particle Nature of Light: The Photon" section explores the idea that light behaves as discrete packets of energy called photons. This concept helps explain how light interacts with matter in a particle-like manner.

Key Points:

Photon: A photon is a tiny packet of energy that makes up light. Each photon carries a specific amount of energy, which depends on the light's frequency.

Photoelectric Effect: When light hits a metal surface, it can eject electrons from the surface. This effect shows that light's energy comes in these packets or photons, not in a continuous wave.

Energy of a Photon: The energy E of a photon can be calculated using the formula:

$$E = h\nu$$

where h is Planck's constant and ν (nu) is the frequency of the light.

This particle-like behaviour of light is crucial for understanding various physical phenomena and is a key part of the dual nature of radiation and matter.